

SYSTEM USABILITY SCALE (SUS) EVALUATION

Machine Learning-Based Interactive Mobile Application for Early Risk Detection and Midwifery Support
Kumarage M.B.S.I.E - IT22606556
Sri Lanka Institute of Information Technology

1. Evaluation Purpose

This evaluation measured the usability of the proposed maternal risk detection mobile application using the System Usability Scale (SUS). The evaluation assessed whether midwifery and nursing participants could confidently complete key tasks such as vital-parameter entry, machine-learning risk-result interpretation, dashboard navigation, and critical-alert response.

2. Participants and Procedure

The usability evaluation recruited 18 participants from midwifery and nursing programs at SLIIT. The participant group included 14 female and 4 male students, with a mean age of 22.3 years and an age range of 20 to 28 years. Fourteen participants primarily used Sinhala, three primarily used Tamil, and one primarily used English.

Each participant completed a 30-minute supervised session using the application. After the practical tasks, participants completed the ten-item SUS questionnaire in their preferred language and took part in a short debrief interview.

3. SUS Scoring Method

The SUS questionnaire contains 10 items rated on a five-point Likert scale. For positively worded items, the adjusted score is rating minus 1. For negatively worded items, the adjusted score is 5 minus rating. The adjusted item scores are summed and multiplied by 2.5 to produce a usability score out of 100.

4. Overall SUS Result

Measure	Result	Interpretation
Number of participants	18	Prototype-level usability testing sample
Mean SUS score	78.5	Good usability band
Standard deviation	6.2	Moderate variation among participants
Score range	67.5 - 92.5	All scores remained close to or above acceptable usability
Cronbach alpha	0.82	Good internal consistency

The mean SUS score of 78.5 is above the common usability benchmark of 68 and falls within the Good acceptability band. This indicates that participants generally found the application understandable, learnable, and suitable for the intended workflow.

5. SUS Score Breakdown by Participant Group

Participant Group	Mean SUS Score	SD	n
All participants	78.5	6.2	18
Sinhala-language interface users	79.1	6.0	14
Tamil-language interface users	76.7	5.8	3
English-language interface users	80.0	-	1
No prior health app experience	77.2	6.8	10
Prior health app experience	80.3	5.1	8

The Sinhala, Tamil, and English interface conditions all remained within the Good usability range. Participants with prior health-app experience recorded a slightly higher mean score than those without prior health-app experience, suggesting that previous exposure to health-related apps may improve confidence, but the application remained usable for first-time health-app users as well.

6. Key Usability Findings

The highest-rated SUS areas were frequent-use intention, ease of use, integration of functions, quick learnability, and user confidence. These findings suggest that the core workflow was clear and that participants understood how the application supported antenatal risk assessment.

The lowest-rated area related to the system feeling cumbersome. This suggests that a minority of participants experienced some first-time complexity, mainly around the data entry form. Future improvement should focus on simplifying input grouping, improving field guidance, and adding a short onboarding flow.

7. Conclusion

The SUS evaluation shows that the proposed mobile application achieved good usability for the target academic evaluation group. The overall SUS score of 78.5, good internal consistency, and positive participant feedback support the application as an assistive tool for improving speed, confidence, and consistency in antenatal maternal risk assessment. Further validation with practicing midwives in real clinical environments is recommended before clinical deployment.

Source: Extracted and summarized from the final thesis document, including Section 2.7, Section 3.4, Table 3.2, Table 3.5, and Appendix B.